

THE STORY OF AGRICULTURE

FROM HUNTER-GATHERERS AND DOMESTICATION TO CULTIVATION AND GENETIC MANIPULATION

No other activity has made a greater impact on both human society and the environment than agriculture. Its discovery and use first allowed small, previously nomadic, hunter-gathering societies to settle in one place, transform the landscape, form communities, and establish civilizations.

Most archaeologists agree that the earliest plants were domesticated in the Fertile Crescent region of the Middle East around 10,000 BCE. While the fig is thought to be the first truly cultivated food, emmer wheat, barley, lentils, chickpeas, and flax were also common early crops. The first animals to be domesticated were sheep and goats, followed by small breeds of cattle.

On the other side of the world, in what is now South and Central America, squash and corn were being planted, joined by beans; these three plants became known as the “Three Sisters” crops and represented an early knowledge of nutrition: planting them together not only retained soil nutrients but provided essential vitamin and minerals needed for human health.

TOOLS OF THE TRADE

Neolithic farmers used digging sticks—long, flat blades with rounded points—to scrape shallow depressions in the earth, into which they dropped seeds. They also cleared areas of woodland using axes as well as fire to make space for crops and animal enclosures. However, sophisticated farming techniques such as irrigation and large-scale monocropping were practiced in Sumeria as early as 5000 BCE, while in Egypt, farmers made use of plows and sickles, and boasted a range of crops and livestock.

The light scratch plow used by Mediterranean countries dominated farming in Europe until the

Middle Ages, when the heavy horse-drawn plow and a three-field system of crop rotation revolutionized agriculture and greatly increased food supplies. Voyages to Asia and the discovery of the New World in the 15th century had a profound impact on agriculture worldwide, as crops and animals were exchanged between Europe, Asia, and the Americas. This changed agriculture on a scale not seen again until engine-driven farm machinery and mass-production techniques, including the use of chemical fertilizers and pesticides, became the norm in the 20th century.

